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الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

الجامعة العربية المفتوحة - مصر
Arab Open University
القبول والتسجيل
الامتحانات
Arab Republic

# M-MT131
# Discrete Mathematics

## Midterm Examination (MTA)
## Spring Semester 2025/2026

## Date: 14 /Apr./2026

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|---|---|
| **Number of Exam Pages:** 3 pages (including this cover sheet) | **Time Allowed:** 90 minutes |

**Instructions:**

- Total Marks: **30**
- This exam consists of **four questions**.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and** ID on the External Answer booklet.
- **Calculators** are allowed.



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**Q–1: [5×3 marks] Choose the correct answer:**

**1.1** Evaluate the expression:

$$11000 \wedge (01011 \oplus 11011)$$

a) 11011
b) 10000
c) 11000
d) 01011
e) None of the mentioned

**1.2** Find a counterexample for the statement were $U = \{3, 4, 5, 10\}$ is the universal set:

$$"\forall x, x \text{ is } prime"$$

a) 3 is a counterexample.
b) The statement is true; hence no counterexample exists.
c) 4 is the only counterexample.
d) 5 is a counterexample
e) None of the mentioned

**1.3** Which of the following is an equivalent to $(U \cap A) \cap (\emptyset \cup \bar{A})$ :

a) $U$
b) $A$
c) $B$
d) $\emptyset$
e) None of the mentioned

**1.4** In an arithmetic sequence with first term 2 and common difference 5, what is the 10th term?

**a)** 9
**b)** 35
**c)** 47
**d)** 45
**e)** None of the above

**1.5** The greatest common divisor of 28 and 13 is

a) 1
b) 2
c) 3
d) 11
e) None of the mentioned

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Each question in below worth 15 marks. Answer all the following:

**Q–2:**

a) **[9 marks]** Show that $(p \to r) \wedge (q \to r)$ and $(p \vee q) \to r$ are logically equivalent.

b) **[6 marks]** Determine the truth values of the statements where the domain of discourse is the set of real numbers, justify your answers:

i. $\forall x(2x > x)$

ii. $\exists x((x-3)^2 \leq 0)$

**Q-3:**

a) **[2+2+2+2 marks]** Suppose the universal set $U = \{x \in N : x \leq 15\}$, $A = \{x \in U : x \text{ is even}\}$, $B = \{x \in U : x \text{ is a multiple of } 5\}$ and $C = \{x \in U : x > 9\}$. Find

i. List of sets A, B and C

ii. $C \cap A$

iii. $B - (C \cap A)$.

iv. $\bar{C}$

b) **[4 marks]** Suppose $f : R \to Z$ where $(x) = \lfloor x \rfloor$, if $S = \{x : 1 \leq x \leq 6\}$, then find $f(S)$?

c) **[2+2 marks]** A theater is designed so that each new row contains a fixed number of additional seats compared to the previous row. The first row contains 18 seats, and the fifth row contains 34 seats.

Assuming the number of seats follows an arithmetic sequence:

i. Determine the number of seats in the twelfth row.

ii. Write a general formula for the number of seats in the nth row.

d) **[1 mark]** Calculate the following sum: $\sum_{i=1}^{10}(i)$.

**Q-4:**

a) **[5 marks]** Let $A = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$. Find $A \odot B$.

b) **[4 marks]** What is hexadecimal representation of $(1101010110)_2$

c) **[6 marks]** Using the decryption function $f(x) = (x - 6) \bmod 26$, $0 \leq x \leq 25$, decrypt the message "YZUV GZ TUUT".

| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

**End of questions**



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